



42299

Human Computer Interaction

2025–2026, 2º semester

Project Logbook

Group 309

[EcoSpot]



Contents

Contents	1
Team Members	2
Task 01: General Idea Description	4
Competitors.....	4
Task 02: User Characterization.....	4
Personas	5
Task 03: Characterization of Use (Scenarios).....	6
Scenarios	7
Task 04a: Requirements.....	8
Task 04b: Consolidated Idea	10
Task 05: Low-Fidelity Mockup — The Paper Prototype.....	11
Task 06: Usability Testing Preparation	16
Methods	17
List of Tasks	17
Consent Form	17
User Table.....	17
Observer Table	18
Post-Tasks Questionnaire.....	19
Task 07: Prototype Evaluation	19
Participants.....	20
Evaluation Results	20
○ Post-task Questionnaire Results (Table 1)	20
○ SUS Questionnaire Results (Table 2)	21
○ Observer Results During Task Execution (Table 3).....	21
Evaluation Discussion	21
Task 08: Refined Mockup	22
Task 09: Critical Analysis of the Project	25
Strengths	26
Weaknesses.....	26
Workload	26
Self-Evaluation.....	28



Team Identification

Team Name	<i>EcoSpot</i>
-----------	----------------

Team Members

Member 1	
Name	<i>Clara Guimarães Vieira dos Santos</i>
NMEC	<i>125006</i>
Course	<i>Aerospace Engineering</i>
Bio	I'm passionate about recovery systems, aerospace vehicles, and the development of optimized engineering solutions. Some of my competences include analytical thinking, teamwork, problem-solving, and participation in drone competitions.

Member 2	
Name	<i>Inês Veigas Cardoso</i>
NMEC	<i>125738</i>
Course	Computer and Informatics Engineering
Bio	I'm interested in software development and cybersecurity. My competences are programming, problem-solving, software development, and teamwork.

Member 3	
Name	<i>Lara Rodrigues Vasconcelos</i>
NMEC	<i>119769</i>
Course	<i>Biomedical Engineering</i>
Bio	I'm interested in medical technology, healthcare innovation, and improving the quality of life for the general population. My competences are biomedical device design, problem-solving, data analysis, and teamwork.



Member 4	
Name	<i>Joana Castro Silva</i>
NMEC	<i>119844</i>
Course	<i>Computers and Informatics Engineering</i>
Bio	<i>I'm interested in data analytics and software development. My competences are attention to detail, programming, problem-solving and teamwork</i>

Member 5	
Name	<i>Francisco Ladeiro Belo dos Santos</i>
NMEC	<i>126255</i>
Course	<i>Computer and Informatics Engineering</i>
Bio	<i>Software developer, passionate about building secure and intelligent systems. I leverage my skills in programming, problem-solving, and teamwork to tackle complex challenges at the intersection of Artificial Intelligence and Cybersecurity.</i>

Member 6	
Name	<i>João Pedro Ramos Vitória da Silva</i>
NMEC	<i>115879</i>
Course	<i>Computer and Informatics Engineering</i>
Bio	<i>Aspiring Cybersecurity Specialist with a strong foundation in Python. I focus on writing clean, tefficient code to automate security processes, combining analytical thinking with a drive to protect and secure digital environments.</i>

Member 7	
Name	<i>Nelson Ramos</i>
NMEC	<i>124921</i>
Course	<i>Computer and Informatics Engineering</i>
Bio	<i>Computer and Informatics Engineering student at University of Aveiro, passionate about systems programming and infrastructure engineering. My competences include DevOps/SRE practices, cloud infrastructure, and low-level C development, with hands-on experience building production-grade systems from scratch. Main experience is Cloud Architecture (I am a Certified Solutions Architect (AWS's SAA-C03 Certification)).</i>



Task 01: General Idea Description

With this project, our goal was to develop a human-centered technology system to help maintain our planet's health in the present and near future. Our system's focus is to transform the way people see the habit of recycling: instead of a boring, complex, and also time-consuming task someone has to do out of obligation, it becomes a fun, rewarding, and easy part of the user's daily routine.

The system was designed for a mobile app in which users will be able to quickly locate nearby recycling bins, identify the correct recycling bin for different materials, and check the availability of recycling machines that exchange recyclable materials, such as plastic bottles, for money. It was designed to support users with different levels of environmental awareness, going from users who are already committed to sustainability to individuals who currently recycle very little, or don't recycle at all.

Competitors

There are several competing systems related to recycling that are already on the market (for example: Wasteapp and Trash4Good), mainly in the form of mobile apps, but they are only focused on recycling bin location services. Despite it being very useful, the existing systems tend to focus only on the location part, and they don't provide information about the recycling of unusual materials, or even common ones. Normally, these systems are used by people who are active in the recycling process.

These systems have several strengths, including making it easy to access nearby recycling points, but they also have limited functionality and lack of interaction with the user. As a result of all this, users often need to rely on multiple platforms to access different recycling-related services to get all the information they need to properly do it and often get discouraged.



Task 02: User Characterization

We chose these personas because they represent the diverse recycling mindsets and behaviors for our target audience. They capture the diversity of our potential users: from students who already recycle and seek a more efficient way to do it, to those who do not currently recycle but might be motivated to start. By acknowledging these different levels of experience, these personas allow us to design an app that effectively meets different needs and challenges.

Personas

Manuel, 18 years old, University student



Manuel is a 1st year university student, currently relocating to a different city to study. Therefore, he does not yet know where, near his residence, he can go to properly recycle his waste.

Manuel is tech-savvy, in that he has been using technology, namely phone apps, his whole life.

His days typically revolve around his studies, and not much else: he gets up, goes to class, dedicates some after-class time to revising the materials. In the evening, he cooks for himself and goes to sleep.

Manuel aims to lead a cleaner, more sustainable life and contribute to a better ecological environment tomorrow.

He often gets frustrated with the lack of resources, on campus and near his residence, for proper recycling.

Motivation: As much as Manuel struggles to find where he can recycle his waste, using our system will enable him to promptly find where the available recycling points are.

Paula, 34 years old, Working Mother



Paula is a working mother from Porto, living with her husband and two children. She is always very busy due to work and her family responsibilities, which leads to her struggling to find time to properly recycle. She tries to teach her children environmentally friendly habits, but sometimes she is unsure where to put certain materials or where the recycling machines are in Porto.

She enjoys practical and fast solutions that make daily tasks easier and help her family to contribute to a more sustainable future.

Paula is constantly working on her computer and phone, so she is very familiar with modern technology and would appreciate an app that can help her recycle. Paula believes recycling is important not only for adults but for children. She gets really frustrated when the recycling bins are really far away or not available.

Motivation: Paula has limited time, and by using the system it would help her quickly locate nearby recycling bins and machines, making recycling easier for her and her family. She likes to quiz her children with recycling terms and concepts.



Carlos, 68 years old, Retired Citizen



Carlos is a retired man living in Lisbon who cares about keeping his neighborhood clean and protecting the environment around him. Although he normally separates household waste, he sometimes feels confused about which recycling bin should be used for uncommon materials, like perfume bottles, old pots... He also gets frustrated when the information given on the recycling bin is unclear or difficult to understand.

Carlos is not very comfortable using technology and normally prefers simple and intuitive systems.

In his daily routine, Carlos enjoys taking morning walks around the neighborhood, reading the newspaper, and occasionally helping at the local parish.

He has basic literacy but no experience with smartphones, relying on his family for any technology-related tasks.

Motivation: Since Carlos sometimes has difficulty correctly separating waste, using this system would help him easily identify the correct recycling bins and nearby recycling locations near him.

Inês, 30 years old, Sales Assistant



Inês works as a sales assistant in a shopping center near her house where she lives alone. She has never really paid attention to recycling because of her busy lifestyle and often sees it as an inconvenience and a time-consuming thing that won't change anything.

Usually, she throws away most packages without separating the materials because she doesn't even know where the ecopoints are located.

Inês is familiar with mobile apps and is driven by convenience and incentives. Inês has tried to recycle before, but often gave up, as she struggled to find nearby recycling bins and was unsure where to dispose of certain materials. She tried a few times and failed, and eventually gave up recycling, and went back to easier habits.

In her daily routine, Inês commutes to work, spends long shifts on her feet at the shopping center, and unwinds at home in the evenings, usually scrolling through social media. She is comfortable with apps for entertainment and shopping, but has never used any app with an environmental or civic purpose.

Motivation: She is someone that values convenience and incentives; the system would encourage her to recycle more by making the process easier, faster, and financially rewarding.



Task 03: Characterization of Use (Scenarios)

To reach these scenarios we brainstormed and discussed common recycling difficulties faced by different types of users. We based the scenarios on the personas previously created, focusing on their daily routines, motivations, and frustrations.

Scenarios

Scenario 1: Finding Nearby Ecopoints After Moving

In his apartment, Manuel cooked dinner and realized he had collected enough plastic packaging, paper boxes, and glass bottles. He didn't know where to dispose of them properly, as he had just moved to a new city and didn't know the nearest recycling points.

With the app, Manuel can quickly find recycling bins close to his home. The system displays the nearest recycling points on a map and tells him what materials can be deposited in each bin. Thanks to the app Manuel can recycle correctly, without wasting time looking around the city.

Scenario 2: Recycling on Campus Between Classes

In university, Manuel has a break between classes and decides to buy lunch at the cafeteria and consumes a soft drink and a snack. After finishing, he realizes he has plastic and a soda can that could be recycled. Since he has recently moved to the city, he is not sure where the nearest recycling machine is located and if it is currently available. Before entering his next class, Manuel opens the app to check if there are nearby recycling machines on campus, to see if he can earn a little bit more money. The system quickly shows the closest available machine, its location, and which materials are accepted in it.

Following the suggested machine on the map, Manuel deposits the soda can into the machine and receives a small financial reward and some points in the system. Although the amount is small, it helps him save some extra money as a university student while making recycling easier. This will encourage Manuel to continue maintaining the sustainable lifestyle he values without wasting time during his busy day.

Scenario 3: Recycling During a Busy Family Routine

Paula tends to keep a lot of materials at home over the week such as plastic bottles, cardboard, glass containers and used batteries until she has time to dispose of them properly. She is often too busy between work and taking care of her children to get around to the task.

At least once a week, Paula turns recycling into a small family activity, like a quiz. Before disposing all the materials, she asks her children to guess which recycling bin each item belongs to. Using the app's search feature, they check together where each material should be deposited and learn the correct recycling method, without it being boring. By transforming recycling into a quiz, Paula helps her children develop environmental responsibility in a fun and educational way, while also making household recycling more organized.



Scenario 4: Identifying the Correct Eco point for Uncommon Waste

Carlos wants to dispose of some old perfume bottles and a damaged cooking pot, but he is uncertain about which recycling bin should be used for these materials. The information that is written on nearby recycling bins is confusing and difficult for him to understand.

By using the app, Carlos searches for how to recycle some items. The system explains in simple terms how each object should be separated and shows the nearest suitable recycling locations. With the app's simple and intuitive interface, Carlos feels even more confident about recycling correctly and contributing to a cleaner neighborhood.

Scenario 5: Recycling Motivated by Convenience and Rewards

After Inês finished work at the shopping center, she bought a drink and noticed a recycling machine that trades materials into money, near the supermarket entrance. Normally, she would throw the bottle away because recycling felt inconvenient to her. However, she opened the app and discovered that the machine accepts plastic bottles and provides small financial rewards, and the app gives an award too in return. The app also shows additional nearby recycling bins for other recyclable materials. Motivated by the convenience and incentives, Inês started separating more waste at home and recycling more frequently.

In a few weeks, Inês began feeling less motivated because she believed that her individual contribution did not make much difference in the world. However, one day she received a reminder notification from the app showing the positive environmental impact of her recycling. At the same time, the app's ranking feature motivated her to continue, because she noticed she was behind her group of friends in recycling points, and competitive as she is, that wasn't an option. She wanted to continue her progress, so she decided to continue recycling, turning it into a consistent habit rather than a temporary effort.

Scenario 6: Using Voice Assistance to Find an Ecopoint

After cleaning her house, Ines left her apartment carrying some heavy recycling bags filled with plastic bottles, glass containers, and cardboard boxes. Since both of her hands were occupied, she couldn't easily use her phone to search for nearby recycling bins.

Using the app's voice assistance feature, Ines asked the system to locate the nearest recycling point. The app responded with voice directions and guided her to the closest recycling bin near her street. With the hands-free functionality, Ines could recycle her waste conveniently and safely without needing to stop or handle her phone while carrying the bags.



Task 04a: Requirements

○ Functional Requirements

Priority	ID	Requirement
1	R01 (Scenario 1)	Allow the user to search for a nearby recycling bin, based on their location.
1	R02 (Scenario 1)	Display the recycling bins on an interactive map.
1	R03 (Scenario 1)	Give information about what materials can be disposed of at each recycling bin.
2	R04 (Scenario 1,6)	The recycling bin must appear by the user's distance.
2	R05 (Scenario 1)	Provide directions to the selected recycling bin.
1	R06 (Scenario 2)	Allow the user to search for nearby recycling machines based on the user's location.
1	R07 (Scenario 2)	Display the availability status of the machine.
1	R08 (Scenario 2,5)	The system should give information on which material is accepted in that machine.
3	R09 (Scenario 2)	Inform users of financial rewards or points received on the recycling machines.
1	R10 (Scenario 2,5)	Encourage sustainable habits through points and rewards.
1	R11 (Scenario 3)	The system should allow users to search where each material should be disposed of.
2	R12 (Scenario 3)	Have a section on which you can quiz yourself, more like a game to teach kids how to recycle.
1	R13 (Scenario 4)	Search where to recycle uncommon items.
2	R14 (Scenario 4)	Provide clear instructions on how each item should be disposed of.
2	R15 (Scenario 5)	The system includes a ranking that can be used with friends.
3	R16 (Scenario 5)	Send reminder notifications to encourage recycling habits.
1	R17 (Scenario 6)	Search for nearby recycling bins by using voice commands.
1	R18 (Scenario 6)	The system provides voice guidance until the recycling bin.



○ Non-functional Requirements

Priority	ID	Requirement
1	R01 (Scenario 4)	Simple and intuitive interface, to be accessible by users with low technological experience.
1	R02 (Scenario 2,3)	The system should be responsive to support quick access to busy routines.
1	R03 (All Scenarios)	System adapted for mobile devices.
1	R04 (Scenario 3, 4)	The information given should be clear and the language simple.
1	R05 (Scenario 6)	The voice interaction should be accurate to help with hands-free usage.



Task 04b: Consolidated Idea

Aspect	Description
Platform	<i>Cross-platform (Android and iOS) - It can reach most users and will have easy access.</i>
Interaction	<i>Touch and Voice - They can use normal navigation, but also voice commands, for situations like carrying a lot of recycling bags.</i>
Device	<i>Smartphone - It is easier for GPS location and mobility, but also more convenient (possibility of using it anywhere).</i>



Task 05: Low-Fidelity Mockup — The Paper Prototype

For the development of the mockup, the Figma platform was used, through which the system can be accessed via the following link: <https://www.figma.com/make/2najsOozPxb6iHd13L1cXK/Sustainable-Recycling-App?t=WifkjFpSgl140CdQ-1>. Figma's AI section was also used, and the following prompts were applied to achieve the final result:

- “With this project, we aim to develop a human-centered technological system to help preserve our planet's health. Our system's main goal is to transform the way the user sees the habit of recycling: instead of a mere boring, complex, time-consuming task someone must do out of obligation, it becomes a tangible, rewarding, and easy part of the user's daily routine. We want to influence our users to have a more sustainable and ecological lifestyle. With our system, our users will be able to reduce their ecological footprint through positive reinforcement.”
- “Add a tab that allows users to search where to dispose of a specific material by identifying the appropriate recycling bin, as well as the location of nearby recycling bins. Additionally, include recycling machines that provide monetary rewards, displaying their location and whether they are available or out of service.”
- “But add a map instead of a search bar to find recycling bins and recycling machines.”
- “I also want you to add a search section to help users know where to dispose of each material, allowing them to search for a material and identify the correct recycling bin. Additionally, I think the sections displaying progress and rewards could be combined into one, or rewards could be integrated with the impact section.”
- “I want there to be a section in the search tab for what I mentioned about searching materials, where users can search for a material and find the correct recycling bin. I also want a separate map tab showing the location of nearby recycling machines and recycling bins. In that section, a search option should also be included so users can search for a specific place or location.”
- “In the map section, include a GPS-style map image of Aveiro showing the location of all existing recycling bins and recycling machines.”
- “In the materials guide section, include a quiz with recycling-related questions and answers, which can be presented either as multiple-choice questions or direct-answer questions to make the experience more interactive and educational. For the map section, improve the GPS functionality because it freezes when opened. Instead of loading a heavy live GPS map immediately, use a lighter and more optimized map interface that displays the locations of nearby recycling bins and recycling machines without causing performance issues.”

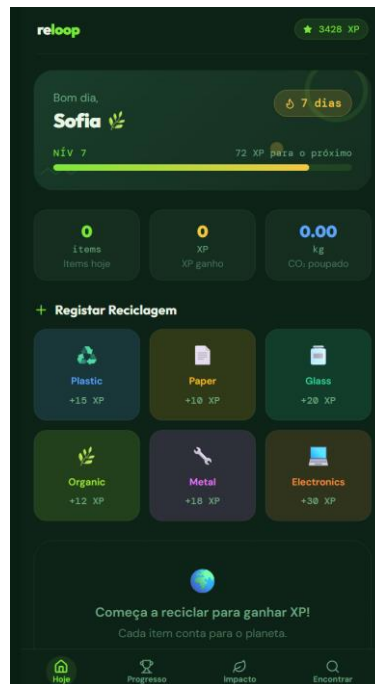


Figure 1 - Main screen of the system, in which the user can register the materials recycled and the overall contributions.



Figure 2 -The progress section that allows the user to know the level of recycling and the weekly activity.

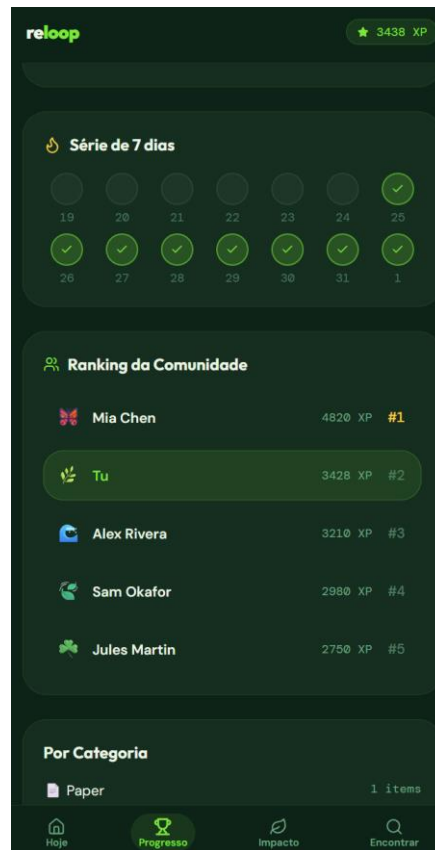


Figure 3 - In the same section as in figure 2, the user can view the streak and compare the ranking with friends.

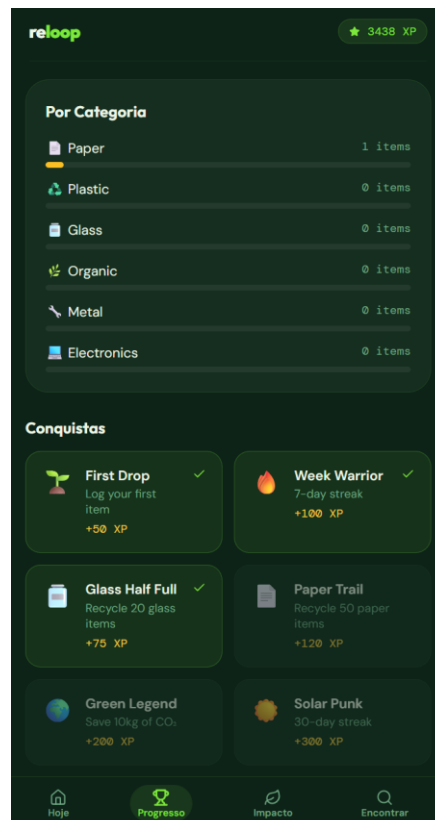


Figure 4 - This screen allows the user to track their progress by category and achievements over time.



Figure 5 - This section shows the impact the user has made on the earth, for example: Carbon dioxide saved, water saved, recycled items...



Figure 6 - This feature helps users identify the correct recycling point for different materials with a search tool and voice command, but also a quick recycling guide.



Figure 7 - In the search section, there is a link to a map with nearby recycling bins and machines.



Task 06: Usability Testing Preparation

Methods

We decided to make our test-users perform some simple tasks (listed in the list of tasks below) and also make them evaluate the usability of our prototype. We passed a consent form as well as the list of tasks and the SUS questionnaire at the end. We chose this option as it is the easiest, quickest, and most realistic way of having real data about the usability of the system we developed. That way, it's easier to understand if there is something we need to refine, delete, or add in order to improve the usability of our system.

List of Tasks

Question	Description
Find the nearest recycling bins on the map.	The user must go to the map and find the nearest recycling bin by distance.
Find in which recycling bin to deposit a certain material (example: metal).	The user must find the search bar and search for the desired material.
Answer the quiz inserted in the app (the right and wrong answers do not count towards evaluation).	The user must find and complete the quiz in the app.
Find a recycling bin using the voice command.	The user must go to the map, find the voice command, and search for the recycling bin.
Find and compare the rankings inserted into the system.	The user must find the ranking in the app and compare it to other users.
Select a material and inform the app that you recycled it.	The user must select a material and click on it in the app.

Consent Form

This form was answered by the users before the test began to get all the approvals and information.

Consent Form (Adult)

I agree to participate in the study conducted by the University of Aveiro.

I understand that participation in this usability study is voluntary and I agree to immediately raise any concerns or areas of discomfort during the session with the study administrator.

Please sign below to indicate that you have read and you understand the information on this form and that any questions you might have about the session have been answered.

Date: _____

Please print your name: _____

Please sign your name: _____

Thank you!

We appreciate your participation.

Figure 8 - Consent Form.



User Table

Teste com utilizadores Exemplo

Teste 1:

O principal objetivo deste teste é avaliar a usabilidade e eficácia de um sistema de reciclagem com o intuito de procura de ecopontos e máquinas de reciclagem próximas, no qual permite aos utilizadores identificar corretamente onde depositar diferentes materiais recicláveis. Pretende-se também avaliar a facilidade de utilização das funcionalidades do sistema, como a localização por mapa, assistência por voz, recompensas e informação sobre reciclagem.

O grupo de utilizadores considerado neste teste inclui estudantes universitários e trabalhadores, uma vez que estes representam diferentes perfis com diferentes necessidades diárias relacionadas com a reciclagem. O sistema deverá ser intuitivo, acessível e prático, para uma utilização simples em diferentes contextos do quotidiano.

Nota: Na utilização do Mapa, o texto apresentado nessa secção é a tradução dos pontos obtidos no mapa. A localização GPS existente no mapa é referente a cidade do Porto e não se encontra visível a localização atual.

Tarefa 1	Encontrar o Ecoponto mais próximo no mapa ----- Nada Fácil 1 2 3 4 5 Muito Fácil
Tarefa 2	Encontrar em que ecoponto depositar certo material (exemplo: Metal) ----- Nada Fácil 1 2 3 4 5 Muito Fácil
Tarefa 3	Resolver o <u>Quiz</u> inserido no sistema (as respostas erradas e certas não contam para avaliação) ----- Nada Fácil 1 2 3 4 5 Muito Fácil
Tarefa 4	Encontrar um Ecoponto a utilizar o comando de voz ----- Nada Fácil 1 2 3 4 5 Muito Fácil
Tarefa 5	Encontrar e comparar o Ranking inserido no sistema ----- Nada Fácil 1 2 3 4 5 Muito Fácil
Tarefa 6	Selecionar um material para informar a aplicação que o reciclaste ----- Nada Fácil 1 2 3 4 5 Muito Fácil

Figure 9 - Table used after each task.



Observer Table

Nº de utilizador: _____

Guião do Observador

Tarefa	Nº cliques	Completo(a) a Tarefa?	Tempo Máximo Tempo observado (mm:ss)	Cometeu erros?	Sentiu-se perdido?	Solicitou ajuda	Grau de facilidade observada 1 – Nada Fácil 5 – Muito Fácil					
1		não ☐ sim ☐	2m _____	não ☐ poucos ☐ muitos ☐	não ☐ pouco ☐ muito ☐	não ☐ sim ☐ qual?	<table border="1"><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr></table>	1	2	3	4	5
1	2	3	4	5								
2		não ☐ sim ☐	2m _____	não ☐ poucos ☐ muitos ☐	não ☐ pouco ☐ muito ☐	não ☐ sim ☐ qual?	<table border="1"><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr></table>	1	2	3	4	5
1	2	3	4	5								
3		não ☐ sim ☐	2m _____	não ☐ poucos ☐ muitos ☐	não ☐ pouco ☐ muito ☐	não ☐ sim ☐ qual?	<table border="1"><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr></table>	1	2	3	4	5
1	2	3	4	5								
4		não ☐ sim ☐	2m _____	não ☐ poucos ☐ muitos ☐	não ☐ pouco ☐ muito ☐	não ☐ sim ☐ qual?	<table border="1"><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr></table>	1	2	3	4	5
1	2	3	4	5								
5		não ☐ sim ☐	2m _____	não ☐ poucos ☐ muitos ☐	não ☐ pouco ☐ muito ☐	não ☐ sim ☐ qual?	<table border="1"><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr></table>	1	2	3	4	5
1	2	3	4	5								

Observações

Figure 10 -Table answered by the observer while the task was being executed.

Post-Tasks Questionnaire

Please enter your participant number: _____

System Usability Scale (SUS)

This is a standard questionnaire that measures the overall usability of a system. Please select the answer that best expresses how you feel about each statement after using the website today.

	Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree
1. I think I would like to use this tool frequently.	☐	☐	☐	☐	☐
2. I found the tool unnecessarily complex.	☐	☐	☐	☐	☐
3. I thought the tool was easy to use.	☐	☐	☐	☐	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5. I found the various functions in this tool were well integrated.	☐	☐	☐	☐	☐
6. I thought there was too much inconsistency in this tool.	☐	☐	☐	☐	☐
7. I would imagine that most people would learn to use this tool very quickly.	☐	☐	☐	☐	☐
8. I found the tool very cumbersome to use.	☐	☐	☐	☐	☐
9. I felt very confident using the tool.	☐	☐	☐	☐	☐
10. I needed to learn a lot of things before I could get going with this tool.	☐	☐	☐	☐	☐

How likely are you to recommend this website to others? (please circle your answer)

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

Figure 11 - Questionnaire answered by the user, after all the tasks.



Task 07: Prototype Evaluation

We performed some of the evaluations in class, with the help of some classmates who volunteered to test our prototype. We also did some evaluations outside of the class with people from different perspectives to evaluate whether the prototype fits the majority of our personas.

Participants

We had 7 participants, in an age range between 19 and 50 years old, with 5 female and 1 male participant. They were university students, but also working people, familiar with similar interactive systems.

This mix of profiles was intentional, as it allowed us to cover a range of personas defined for the system: younger tech-savvy users similar to Manuel, busy adults similar to Paula, and less tech-experienced users similar to Carlos. Although none of the participants perfectly matched every persona, their diversity in age, occupation, and technological familiarity made their feedback representative of the target audience.

Evaluation Results

The evaluation included 3 different assessment methods: a questionnaire completed after each task (Table 1), the System Usability Scale (SUS) questionnaire (Table 2), and observer data collected during the tasks (Table 3).

○ Post-task Questionnaire Results (Table 1)

After completing each task, participants were asked to rate how easy or difficult the task was on a scale from 1 (Very Difficult) to 5 (Very Easy). In general, the results were positive, with most tasks receiving ratings above 4, suggesting that participants found the application relatively easy to use.

Tasks 5 and 6 received the highest ratings, indicating that these features were easy to understand and interact with. Another reason for that can be that the user is more used to the system on the last task, as participants appeared comfortable navigating these sections, completing them quickly and without major difficulties.

On the other hand, task 1, which involved finding the nearest recycling bin on the map, received the lowest average score. Although participants were still able to complete the task, some encountered a lot of difficulties, which may be related to the map interface problem, but also the impact of operating a new system. The task 3 obtained in comparison the same average as task 2.

Table 1 - Data evaluation given by the user about each task individually.

	Average - Tasks							
	Evaluation							
	Participant 1	Participant 2	Participant 3	Participant 4	Participant 5	Participant 6	Participant 7	Total:
Task 1	4	5	5	3	3	5	3	4
Task 2	4	4	5	4	5	4	4	4,29
Task 3	4	5	5	4	5	4	3	4,29
Task 4	5	5	5	4	4	5	4	4,57
Task 5	5	5	5	5	5	4	4	4,71
Task 6	4	5	4	4	5	5	5	4,57



○ SUS Questionnaire Results (Table 2)

To better understand users' perception of the system, participants completed the System Usability Scale (SUS) questionnaire after completing all tasks. The scores ranged from 62.5 to 95, which resulted in an average score of 79 points.

This result suggests that users in general perceived the system positively and considered it easy to use. Most participants felt comfortable interacting with the interface and were able to understand the system's functionalities without major difficulties. The SUS results indicate a good level of usability and overall user satisfaction, although there is always room for improvement.

Table 2 - SUS evaluation data completed by the user.

	Average-SUS
	Questionnaire
Participant 1	75
Participant 2	87,5
Participant 3	85
Participant 4	67,5
Participant 5	95
Participant 6	80
Participant 7	62,5

○ Observer Results During Task Execution (Table 3)

While the task was being executed, an observer recorded information such as the number of clicks, task completion time, errors made, whether participants felt lost, and if assistance was required, which was all noted on Table 3.

The results showed that all participants completed every task, which demonstrates that the system was minimally understandable and functional. However, there were some differences between tasks. Task 3, which was a quiz, required the highest number of clicks and the longest completion time, which is normal since they had to answer questions and it takes some time to go through all of them.

Tasks 5 and 6 were completed faster and with fewer errors, reinforcing the positive ratings given by participants on the post-task questionnaire. Additionally, participants normally did not feel lost while using the system and rarely asked for help, suggesting that the application's structure and navigation were intuitive.

Table 3 - Data collected through observer questionnaire.

	Average-Observer									
	Number of clicks	Completed the task?	Time	Errors Committed	Caption - Errors Committed	Felt lost?	Caption - felt lost?	Asked for help?	Caption - Asked for help?	Observed ease level
Task 1	5	Yes	0:44	2	Few	1,71	A little	2	Yes	3,6
Task 2	7	Yes	0:52	1,71	Few	1,57	A little	1,14	No	4
Task 3	21	Yes	1:22	1,14	None	1,57	A little	1,29	No	4,4
Task 4	7	Yes	0:23	1,43	None	1,43	No	1,43	No	4,4
Task 5	5	Yes	0:13	1	None	1,14	No	1	No	4,9
Task 6	6	Yes	0:17	1,14	None	1,14	No	1	No	4,7



Evaluation Discussion

Overall, the evaluation results show that the system provides a positive experience and is effective in supporting recycling-related activities. However, some aspects were recommended by the users, and some were observed. Therefore, there were some tips for improvement:

- A feature to delete the selected material;
- Map should function directly / Map should work directly;
- The recycling point should appear more clearly when a material is searched (pop up on the screen);
- Change the map location/layout;
- Add a score to the quiz;
- Make the Recycling Log more functional;
- Place the quiz in a more visible area.



Task 08: Refined Mockup

After the evaluation, the system was refined on the same platform as mentioned before. Several improvements were implemented to facilitate usability and create a better experience for the user.

The following changes were made:

- A feature to delete the selected material was added, which allows the users to correct mistakes more easily, figure 12
- The map functionality was improved to work directly and provide a better and easier navigation experience, figure 13
- A scoring system was added to the quiz to increase user engagement and gives feedback on their performance, figure 14
- The recycling bin now appears more clearly when a material is searched, making it easier for users to identify where to dispose of them, figure 15.

These refinements were also aligned with Nielsen's usability heuristics, particularly user control and freedom, visibility of system status, recognition rather than recall, and consistency, helping to improve usability and the overall user experience. Only with another round of tests would it be possible to know if the added improvement made the system more accessible and easier to use.

The improvements were also done with the AI's feature of the platform Figma, with the use of these prompts:

- "Some functionalities still need improvement, such as adding an option to remove the selected material in the system's base section and making the "+" recycling registration feature functional for users who wish to use it. The quiz should be better defined and more visible, while also including a final score at the end. When searching for the correct recycling bin for a material, the answer should pop up on the screen to make it easier for the user to identify the correct option. Additionally, the map could be placed in the bottom central section of the navigation panel to provide easier and quicker access for users."

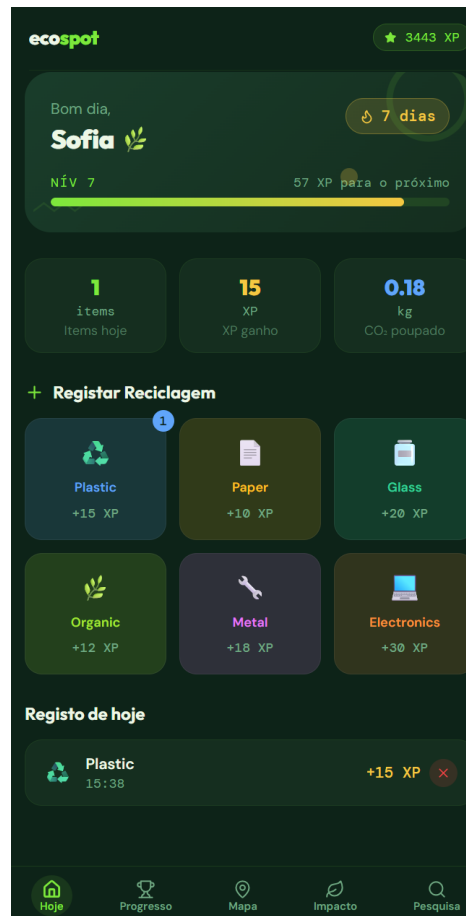


Figure 12 - The new option for deleting the selected material.



Figure 13 - New template for the maps, with the option in the center for easier access.



Figure 14 - The feature for the score at the end of the quiz.



Figure 15 - Section that appears after searching where to put a recyclable material.



Task 09: Critical Analysis of the Project

Strengths

The project presented several strengths throughout the progress in terms of creativity and functionality. The system provides an intuitive interface that helps users identify recyclable materials and locate recycling points. Features added like the material search, map, quiz, and recycling log increased the engagement for the user but also the environmental awareness. Additionally, the usability evaluation showed, in general, positive feedback, and suggested improvements were implemented.

Weaknesses

Some weaknesses were identified during the evaluation process and post-questionnaire. Certain functionalities, such as the map and the quiz, needed some improvement to be more useful and accessible. Future improvements could include additional functionalities and further usability testing with more users, to be able to test new features and overall usability. Not all the requirements were executed, but we considered the most important ones. Also, there were some improvement suggestions that were limited to the platform used for the system.



Workload

N. Mec.	Name	% of Work	Description
125006	Clara Santos	45	I was responsible for structuring the project and its development throughout the weeks of classes. I contributed with part of Task 1, completed the majority of Task 2, and did Tasks 3, 4a, and 5. Additionally, I completed the remaining work for Task 7, as well as Tasks 8 and 9. I was also responsible for editing all the documents used during the evaluation process with the users and data analysis of each result, which were later integrated into Task 7. Additionally, I was fully responsible for the system development on the Figma platform, including the design, structure, and prototype of the interface.
125738	Ines Cardoso	15	I contributed with the identification of the problem, as well as brainstorming the solutions for it. I wrote most scenarios and added the photos on the personas and some texts on them. I was also an assistant through evaluations. I added the consent form, observer table, and post-questionnaire.
119769	Lara Vasconcelos	1	I wrote my short bio on team identification and helped in the scenarios of task 3.
119844	Joana Silva	15	I helped with brainstorming ideas for the system as well as with the introduction of the project (Task 1), with the list of tasks and methods for Task 6 and with the introduction and participants of Task 7. I was also an assistant through some of the evaluations.
126255	Francisco Santos	7	I did tasks 4b and helped with 4a with careful attention to detail and consistency. Conducted a comprehensive review of the project, identifying key areas for refinement and improvement.



			Contributed to the overall quality and coherence of the final project deliverables.
115879	João Silva	10	Did one persona with the help of Nelson and wrote the contents. I also spent some time brainstorming new ideas and solutions for a few problems.
124921	Nelson Ramos	7	Did half of a persona with João, reviewed the document, and made some good suggestions. Also helped with task 7 (participants)

Self-Evaluation

Self-evaluation of the work (and why)	Score: 18/20
In my opinion, this project deserves to be graded 18/20, insofar as it was completed successfully and achieved its main objectives, proposed by the professor. The prototype is functional, easy to use, and includes relevant features to use daily. The user's feedback and evaluation results were considered to improve the system. Nevertheless, some aspects could still be further refined, and additional functionalities could be implemented for the user to have a better experience.	